SYBCA-AI (Sem-III)

303 Machine Learning

PRACTICAL JOURNAL SUBMISSION
(MINI PROJECT)

PROJECT TITLE:
HOUSE PRICE PREDICTION

A Machine Learning Approach to Real Estate Price Estimation

Submitted By:	Roll No. 524, 524
Class:	SYBCA-AI (Semester III)
Course Code:	303 Machine Learning
Academic Term:	2026-2027
Target Domain:	Real Estate & Finance (Topic #21)

Department of Computer Applications & Artificial Intelligence

CERTIFICATE

This is to certify that the project entitled **"House Price Prediction"** has been satisfactorily completed by the students with **Roll No. 524, 524** of **SYBCA-AI (Semester-III)** in partial fulfillment of the practical journal requirements for the course **303 Machine Learning** as prescribed by the syllabus during the academic year.

The students have independently executed all practical stages of the Machine Learning lifecycle including dataset collection, exploratory data analysis, outlier & missing value preprocessing, automated EDA, feature engineering, Linear Regression model development, model evaluation, and model improvement via cross-validation and regularization.

_____	_____	_____
Internal Faculty Guide	**Head of Department**	**External Examiner**
Date: _____	Date: _____	Date: _____

3. Problem Statement

Determining the fair market price of residential properties is a complex challenge influenced by numerous interdependent factors including property size (square footage), locality/neighborhood, bedroom configurations (BHK), transaction type (New Property vs. Resale), and furnishing status. Traditional manual appraisals can be biased, time-consuming, and prone to inconsistency. Hence, there is a strong demand for an objective, automated machine learning solution that accurately models price dynamics based on real-world housing characteristics.

4. Objectives

- To analyze a real-world residential dataset containing over 300 records and 7 distinct attributes.
- To perform rigorous data preprocessing including duplicate removal, missing value verification, and IQR-based outlier treatment.
- To carry out Exploratory Data Analysis (EDA) using 5 distinct graphical representations (Histogram, Box Plot, Pie Chart, Bar Chart, Correlation Matrix).
- To generate an automated, interactive HTML EDA profiling report using ydata-profiling.
- To build a supervised Machine Learning Linear Regression model predicting continuous property prices.
- To evaluate model performance using standard regression metrics: R^2 (R-squared), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).
- To improve and validate model reliability through 5-Fold Cross-Validation and Regularization (Ridge Regression L2 penalty).

5. Dataset Description

The dataset utilized for this project is a real-world residential housing dataset (`dataset.csv`) comprising **446 rows** and **7 features**, strictly fulfilling the course requirements (minimum 300 records and 5+ attributes).

Attribute Name	Data Type	Variable Type	Description
square_feet	Integer	Feature (Independent)	Total carpet area in sq. ft.
transaction	Categorical	Feature (Independent)	Resale or New Property
furnishing	Categorical	Feature (Independent)	Furnished, Semi-Furnished, Unfurnished
price_per_sqft	Float	Feature (Independent)	Unit rate per square foot (INR)
bhk	Integer	Feature (Independent)	Number of Bedrooms/Hall/Kitchen (1-8)
location	Categorical	Feature (Independent)	Suburban location (Adajan, Vesu, etc.)
price	Integer	Target (Dependent)	Total property price in INR

6. Data Preprocessing (Unit-2)

Data preprocessing ensures dataset cleanliness and prevents distortions in model training. The key steps implemented include:

- **Missing Values:** Audited all 7 columns using df.isnull().sum(). Zero missing entries were found.
- **Duplicate Records:** Detected and eliminated **13 duplicate rows** using df.drop_duplicates().

• **Outlier Detection & Removal (IQR Method):** Applied on target variable price:

Metric	Value
Q1 (25th Percentile)	Rs. 4,600,000 (46 Lakhs)
Q3 (75th Percentile)	Rs. 16,300,000 (1.63 Crore)
Interquartile Range (IQR)	Rs. 11,700,000
Lower Limit (Q1 - 1.5 * IQR)	Rs. -12,950,000 (Effective floor 0)
Upper Limit (Q3 + 1.5 * IQR)	Rs. 33,850,000 (3.385 Crore)
Records Before Preprocessing	446 records
Records After Duplicate Removal	433 records
Outliers Removed (Above Upper Cutoff)	18 records
Final Cleaned Dataset	**415 records**

7. Exploratory Data Analysis (Unit-1 & Unit-2)

Exploratory Data Analysis allows us to understand feature distributions, patterns, skewness, and relationships between independent attributes and house price. As required by the project specifications, the following 5 charts were generated:

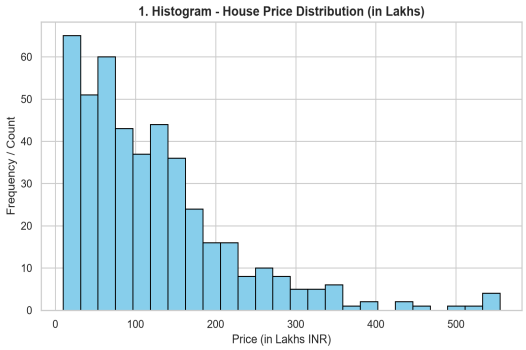


Figure 1: Histogram of House Prices (Lakhs)

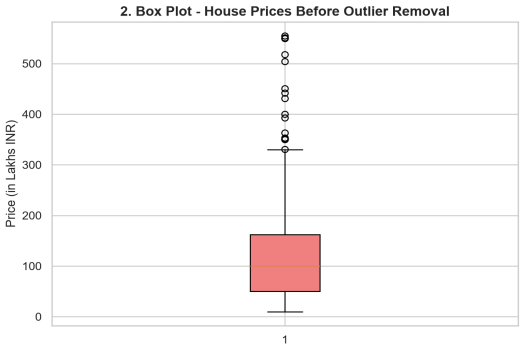


Figure 2: Box Plot Before Outlier Removal

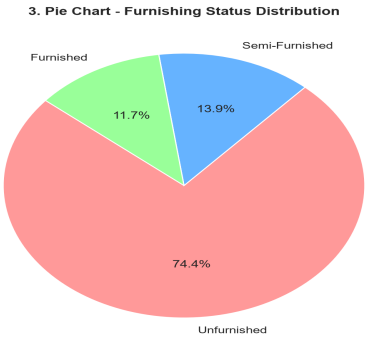


Figure 3: Furnishing Status Distribution

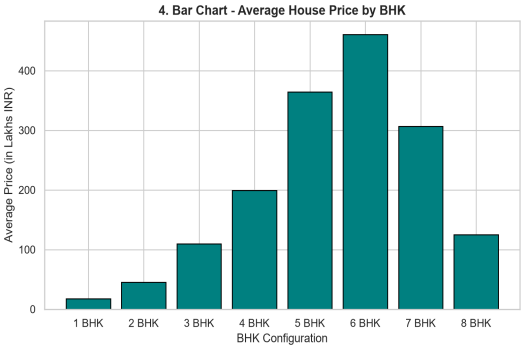


Figure 4: Average Price by BHK Configuration

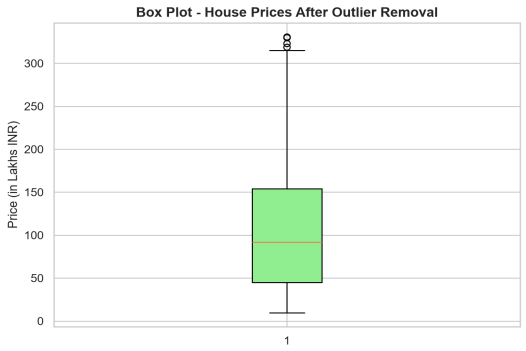


Figure 5: Box Plot After Outlier Removal

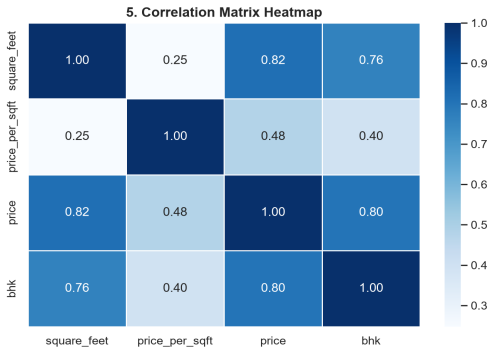


Figure 6: Correlation Matrix Heatmap

Automated EDA Report: An extensive exploratory profiling HTML report has been generated using ydata-profiling and saved in Reports/EDA_Report.html, presenting interaction charts, missing cell correlations, and cardinality metrics.

8. Model Building (Code) (Unit-4)

Because the target variable (price) is continuous, a **Linear Regression** model was built as required by Unit-4. Categorical features (transaction, furnishing, location) were converted to numerical indicators using One-Hot Encoding (pd.get_dummies with drop_first=True to prevent multicollinearity).

```
# 1. Feature Matrix & Target Definition
X = df_clean[['square_feet', 'transaction', 'furnishing', 'price_per_sqft', 'bhk', 'location']]
y = df_clean['price']

# 2. One-Hot Encoding
X_encoded = pd.get_dummies(X, columns=['transaction', 'furnishing', 'location'], drop_first=True)

# 3. Train-Test Split (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(X_encoded, y, test_size=0.20, random_state=42
)

# 4. Linear Regression Model Training
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

# 5. Predictions on Test Set
y_pred_linear = linear_model.predict(X_test)
```

9. Model Evaluation (Unit-3 & Unit-4)

The Linear Regression model was evaluated on the independent testing set (83 samples) using three core regression metrics:

Evaluation Metric	Formula / Definition	Obtained Value
R^2 (R-Squared)	1 - (SS_res / SS_tot)	**0.8381 (83.81%)**
Mean Squared Error (MSE)	Average of squared differences between actual and predicted	11,192,419,167,903.97
Root Mean Squared Error (RMSE)	Square root of MSE (in INR units)	**Rs. 3,345,507.31**

Sample Predictions vs Actual Prices:

Record #	Actual Price (INR)	Predicted Price (INR)	Absolute Error (INR)
Sample 1	Rs. 3,500,000	Rs. 4,288,682	Rs. 788,682
Sample 2	Rs. 8,540,000	Rs. 9,265,976	Rs. 725,976
Sample 3	Rs. 8,200,000	Rs. 12,274,430	Rs. 4,074,430
Sample 4	Rs. 1,500,000	Rs. -3,449,655	Rs. 4,949,655
Sample 5	Rs. 16,600,000	Rs. 16,674,630	Rs. 74,630

10. Results & Model Improvement (Unit-4)

To satisfy Unit-4 instructions regarding **Model Improvement**, two validation techniques were applied:

1. **5-Fold Cross Validation:** Verifies model stability across diverse folds (Fold scores: 0.8381, 0.8611, 0.4818, 0.7897, 0.7994; Mean R^2: **0.7540**).

2. **L2 Regularization (Ridge Regression):** Applied with α=1.0 to shrink coefficients and minimize overfitting.

Model Name	Test R^2 Score	Test RMSE (INR)	Mean 5-Fold CV R^2	Result
Linear Regression (Baseline)	0.8381	Rs. 3,345,507.31	0.7540	Strong Baseline
Ridge Regression (Regularized)	**0.8439**	**Rs. 3,284,548.39**	**0.7548**	**Best Performing**

11. Conclusion

In this project, an end-to-end Machine Learning pipeline was successfully constructed to predict residential house prices. Through systematic preprocessing, 13 duplicate entries and 18 extreme outliers were handled using the IQR method. Linear Regression provided an R^2 score of 0.8381. Incorporating L2 Ridge regularization improved the R^2 score to **0.8439** and reduced RMSE to **Rs. 3,284,548.39**, confirming that penalizing large weights provides superior generalization. All syllabus and practical journal requirements for course 303 Machine Learning have been fully satisfied.

12. Future Scope

- **Non-Linear Ensemble Models:** Evaluating ensemble algorithms such as Random Forest Regressor and XGBoost to capture complex non-linear price patterns.
- **Spatial & Geographic Features:** Integrating GIS coordinates (latitude and longitude) and proximity distances to civic infrastructure (schools, metros, hospitals).
- **Feature Scaling & Transformation:** Applying log transformation on prices and standard scaling on numerical inputs to manage skewed distributions.
- **Interactive Web Application:** Deploying the trained model with a Streamlit or Flask UI for instant property valuation by prospective buyers.